F18CE2 MV Calculus and Real Analysis B: Tutorial 8

1 Define what is meant by $\sum_{r=1}^{\infty} a_n$ converges.

Supose that (x_n) is a convergent sequence and let $a_n = x_n - x_{n+1}$.

Prove that $\sum_{r=1}^{\infty} a_r$ converges and find its sum.

2 Use the identity $\ln(1+1/r) = \ln(r+1) - \ln r$ to find an explicit expression for $\sum_{r=1}^{n} \ln(1+1/r)$.

Hence show that the series $\sum_{r=1}^{\infty} \ln(1+1/r)$ diverges.

3 Prove that $\sum_{r=1}^{\infty} a_r$ diverges when

(a) $a_r = (-1)^r$,

(b) $a_r = \dfrac{2r+1}{r+3}$.

4 State the Comparison Test.

Suppose that $a_r \geqslant 0$ and that the series $\sum_{r=1}^{\infty} a_r$ converges.

Prove that the series $\sum_{r=1}^{\infty} \dfrac{a_r}{1+a_r}$ converges.

5 Discuss the convergence of $\sum_{r=1}^{\infty} a_r$ when

(a) $a_r = \dfrac{r^2}{r^3+1}$,

(b) $a_r = \dfrac{r}{r^3+1}$,

(c) $a_r = \dfrac{(r+1)\sin^2 r}{r^3+3r+1}$.

6 Suppose that the series $\sum_{r=1}^{\infty} a_r^2$ and $\sum_{r=1}^{\infty} b_r^2$ both converge, where $a_r \geqslant 0$ and $b_r \geqslant 0$ for all $r \geqslant 1$.

Show that the series $\sum_{r=1}^{\infty} a_r b_r$ is convergent. [**Hint.** $(a_r - b_r)^2 \geqslant 0$.]

Give an example with $\sum_{r=1}^{\infty} a_r^2$ convergent and $\sum_{r=1}^{\infty} a_r$ divergent.

7 Let (a_r) be a sequence with $Ar^{-1} \leqslant a_r \leqslant Br^{-1}$ for all r where A, B are constants with $0 < A < B$.

Discuss the convergence of

$$(a) \qquad \sum_{r=1}^{\infty} a_r\,, \qquad (b) \qquad \sum_{r=1}^{\infty} a_r^2\,, \qquad (c) \qquad \sum_{r=1}^{\infty} r^{-1}\sqrt{a_r}\,.$$

8 State the Ratio Test. Discuss the convergence of $\displaystyle\sum_{r=1}^{\infty} a_r$ when

$$(a) \qquad a_r = \frac{6^r}{r!}\,, \qquad (b) \qquad a_r = \frac{r3^r}{4^r}\,, \qquad (c) \qquad a_r = \frac{5^r}{4^r\, r}\,.$$

9 Discuss the convergence of $\displaystyle\sum_{r=1}^{\infty} a_r$ when

$$(a) \qquad a_r = \frac{\cos r}{r^2+1}\,, \qquad (b) \qquad a_r = \frac{r^2 \sin r}{r^4+3r^2+1}\,, \qquad (c) \quad a_r = \frac{r^2\,(-1)^r}{2^r}\,.$$

10 Find the radius of convergence R of the following power series:

$$(a) \qquad \sum_{r=1}^{\infty} \frac{(-1)^r r^3 x^r}{3^r}\,, \qquad (b) \qquad \sum_{r=1}^{\infty} \frac{x^r}{6^r\sqrt{r^2+2r}}\,.$$

For part (b), determine also whether the series converges for $x = -R$ and $x = R$.